Portfolio risk formula review

13/11/2026 | Research snapshot in ILS | 100 variants in each of 14 scenario families

The current score is mathematically reproducible, but structurally unreliable. The same assets and cash can score 31 or 17 merely by changing how position rows are represented. Use the stress-loss approach as the leading research candidate, and fix account coverage before presenting a whole-wealth score.

Scope	Reconciled value	Assessment
Investment portfolio	₪1,042,717.80	31 positions; 4 funded accounts; current model 31/100
Separate pension / savings	₪3,202,128.89	8 balances in 3 accounts; allocations unavailable
Combined excluding employee equity	₪4,244,846.69	Balance sum only; whole-wealth risk unavailable

Three leading candidates, with the current formula included

Formula	Quality /100	Portfolio score /100	Financial meaning
1. F2: maximum stress loss	93	52	25.76% loss under the worst chosen illustrative shock
2. F3: assumed factor risk	91	19	9.43% assumed annual risk proxy; uncalibrated
3. F4: 70/30 hybrid	90	42	Blend of F2 and F3; exploratory composite
F1: repaired heuristic	85	16	Smallest model repair; still a heuristic
F0: current v2	54.75	31	Weighted heuristic with row/account sensitivity

These are two different scores. Quality grades measure fit to the declared review rubric. Portfolio scores are model-specific scales. A lower score under a different formula does not mean the portfolio became safer. None of the quality grades measures forecast accuracy.

What was actually executed

1,400 deterministic portfolio variants × 5 formulas = **7,000 formula evaluations**, plus baselines and boundary examples. Each formula received 600 structural comparisons. The variants are sensitivity sweeps and representation checks, not independent market outcomes or a historical backtest.

Recommended next model design: show stressed loss in ILS and percent, modeled debt by account, and data coverage as separate fields. Keep investment suitability and spending needs separate. This review leaves the application and saved portfolio unchanged.

Account values and coverage

The investment balances reconcile using current ledger quantities/cash and saved dated marks. Negative cash is counted as borrowing; none is recorded in this snapshot. USD is converted once at the saved 12/11/2026 rate of **3.1754 ILS per USD**. These are stored-value checks, not independent broker-statement verification.

Account	Cash in account currency	Securities in ILS	Net value in ILS	F0 risk
ETrade ...1048	USD 6,306.57	₪34,278.44	₪54,304.33	47
Schwab ...5122	USD 5,517.20	₪268,338.37	₪285,857.68	42
Schwab ...2873	USD 24,268.91	₪134,115.15	₪211,178.64	34
Psagot ...7641	ILS 38,819.65	₪452,557.50	₪491,377.15	25
Investment total	₪153,428.33	₪889,289.46	₪1,042,717.80	31

Balances excluded from the holdings risk calculation

Savings account	Saved fund balance	Risk coverage
Phoenix ...5606	₪555,407.00	Allocation and economic FX unavailable
Clal ...5606	₪133,911.89	Allocation and economic FX unavailable
Meitav ...6777	₪2,512,810.00	Allocation and economic FX unavailable

The omitted balances represent **75.44%** of combined wealth excluding employee equity. The investment model therefore covers only **24.56%** of that balance total. Pension funds are not assumed to be cash, collateral, or freely transferable funds.

Freshness and source checks

All 31 current positions match saved analyses in identity, quantity, cost basis, asset class and currency. Every saved market value equals quantity × normalized saved unit price. Cash transaction currencies match their accounts, and no future-dated transactions enter the calculation. The extracted JavaScript matches the deployed file hash.

Material gaps: bond 121899CH9 uses a 31/07/2026 mark (105 days old); Treasury 91282CKD2 uses 06/11/2026; fund 5139324 and UNB use 11/11/2026; other stock marks use 12/11/2026. Separate fund balances date from 07/08/2026 or 15/08/2026. Accrued bond interest and clean/dirty price conventions are not independently verified.

The saved analytics cache has a different revision signature from the current ledger. This review reconstructs a dated research input after checking the positions individually; it does not certify a fresh live API context or a current broker valuation. Employee equity is excluded throughout.

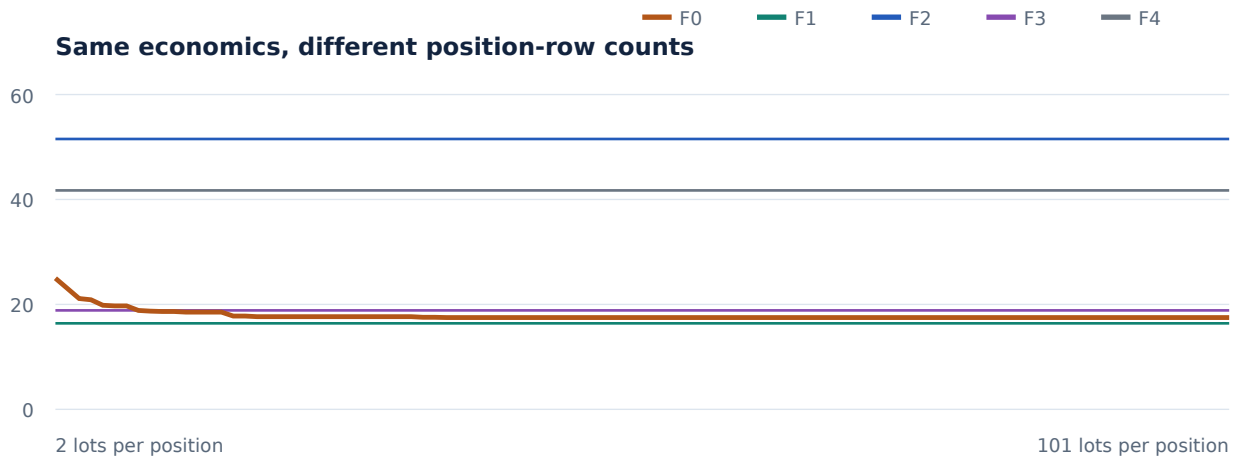
Why the current 31 is questionable

Current v2 is a declared heuristic, not a probability. It sums ten weighted components, amplifies them for borrowing, then blends pooled risk with account risks. The implementation is reproduced exactly in the research bundle.

S = sum of actual component points, with maxima:
concentration 16, asset mix 14, currency 10, industry 10, cash 7, derivatives 10, volatility 10, horizon 10, issuer/country 8, data quality 5
R = round[min(100, S × (1 + D/E))]
Total = round[0.75 × pooled R + 0.25 × gross-asset-weighted account R]

E is signed net equity; D sums each account's negative cash, so positive cash elsewhere cannot erase borrowing. With exposure or debt and E ≤ 0, the score is 100. Empty debt-free portfolios score 0. The current borrowing rule is a useful feature to preserve.

Executed issue	Result	Financial implication
Identical lot splitting	31 → 17; no change in assets or equity	Concentration is partly calculated by row, so representation changes risk.
Equivalent account repartition	Rounded scores range 31-33	The 25% account blend depends on account grouping.
Cash sales at fixed equity	5 reversals in 200 ordered comparisons	Raw risk sometimes rises after moving long exposure to same-currency cash.
Money-market-heavy Psagot account	Current 25; repaired heuristic 3	Wrapper concentration 16 + issuer 4 + long-horizon penalty 3.5 dominate.



The chart uses unrounded research scores; the unchanged baseline F0 is 31 before any split. Proposed models remain constant. A pass/fail comparison uses a 0.0000001-point tolerance; displayed integer plateaus do not hide raw-score movement.

Additional weaknesses: combined positive cash is assigned the base currency, understating the USD-cash component; the price-range proxy is not return volatility; sparse bond marks can create a zero range; and missing pension allocation is absent from the financial score. Excess cash and too little long-term equity are opportunity/suitability concerns and should be reported separately from market loss.

Three leading formulas

1. F2 - maximum illustrative stress loss | quality 93/100

For every selected shock, reprice the local instrument first, then apply its economic FX shock. Keep signed cash and borrowing in their actual currencies. Aggregate the largest nonnegative loss and divide by positive net equity.

$$\Delta E(s) = \Sigma\, V(i) \times [(1 + r(i,s)) \times (1 + f(i,s)) - 1] + \Sigma\, C(a) \times f(a,s)$$
$$L = \max \text{ over } s \text{ of } [\max(0, -\Delta E(s))]$$
$$F2 = \min(100, 200 \times L / E)$$

The scale reaches 100 at a 50% loss under the chosen scenarios. The baseline maximum is **₪268,646.59 (25.76%)**, giving an unrounded score of 51.53, displayed as 52. The shock set is an instantaneous repricing exercise with no assigned probability or forecast horizon. It is not VaR, expected shortfall, or an exhaustive worst case.

Frozen downside shocks: individual equities -50%; broad equities -40%; sector/country equity -50%; multi-asset -25%; money market -1%; government bond -12%; unclassified/IG bond -15%; HY bond -25%; property -40%; commodity -35%. FX paths include ±20% individually and jointly; market paths include both signs. Add individual-stock issuer ±100% scenarios; do not default a diversified fund wrapper as if it were a single stock. All magnitudes are analyst assumptions.

2. F3 - assumed factor risk | quality 91/100

Express each position as ILS exposures to shared equity, rates, FX, and issuer/fund residual factors. Same-issuer residuals aggregate across accounts and lots. Independent factors produce a positive semidefinite covariance structure; their independence and coefficients remain unverified assumptions.

$$\sigma(\text{proxy}) = \text{sqrt}[\Sigma\ \text{factor-ILS-exposure}(k)^2] / E$$
$$F3 = \min(100, 200 \times \sigma(\text{proxy}))$$

Baseline proxy: **9.43%**, giving 18.86, displayed as 19. Common factor assumptions: equity 18%, rates 8%, each foreign currency 12%. Stock residual 30%, government-bond residual 2%, unclassified-bond residual 4%, money-market residual 0.5%; detailed betas are in the frozen methodology. This is not measured historical or forecast annual volatility.

3. F4 - stress / factor hybrid | quality 90/100

$$F4 = 0.70 \times \text{capped } F2 + 0.30 \times \text{capped } F3$$
$$\text{Baseline} = 0.70 \times 51.5281 + 0.30 \times 18.8607 = 41.7279 \rightarrow 42$$

The hybrid combines downside stress and diversification sensitivity, but its 70/30 mix and aligned score scales have no empirical validation. It ranks third under this rubric; prefer F2 and F3 shown separately for decisions. F1, the repaired heuristic, is the smaller implementation option: group issuers, discount fund wrappers, use cash-aware denominators, and remove opportunity-cost penalties and the account blend. Its quality is 85 and baseline score 16.

100 simulations per scenario family

Every family contains exactly 100 deterministic variants; every formula uses the same state in each variant. The table reports the final endpoint score. Detailed raw scores for all 1,400 cases are included in the CSV. Composition changes use frozen marks; no tax, fees, slippage, interest or trading constraints are modeled.

Endpoint - 100 cases each	F0	F1	F2	F3	F4
Sell 100% of USD equities to USD cash	26	10	23	13	20
Redeem 100% of ILS money fund to cash	26	16	51	19	41
Buy equities equal to 50% of initial equity	63	53	100	47	84
Consolidate 100% of stock book into CMRE	38	33	70	28	57
All equities -60%; FX unchanged	29	12	33	12	27
USD/ILS -30%; local prices unchanged	30	14	43	16	35
All bond marks -20%	31	16	51	19	42
Equities -50%, bonds -15%, USD -25%	29	11	31	11	25
CMRE -100%; others unchanged	31	15	48	18	39
External ILS cash equals 100% initial equity	23	8	26	9	21
Assets and cash scaled to 10 times	31	16	52	19	42
101 identical lots per position	17	16	52	19	42
Equivalent USD subaccounts, variant 100	32	16	52	19	42
Assume all omitted funds USD broad equity	49	37	91	37	75

Post-shock risk is not the loss that just occurred. After an equity crash, remaining equity exposure is smaller relative to unchanged ILS cash/money funds. A recalculated score can fall even though the investor has suffered a loss. The P&L below measures that loss directly.

Shock endpoint	Change in value	% initial equity	Remaining equity
All equities -60%; FX unchanged	₪-219,328.46	-21.03%	₪823,389.34
USD/ILS -30%; local prices unchanged	₪-165,402.19	-15.86%	₪877,315.60
All bond marks -20%	₪-14,236.91	-1.37%	₪1,028,480.89
Equities -50%, bonds -15%, USD -25%	₪-282,923.71	-27.13%	₪759,794.09
CMRE -100%; others unchanged	₪-47,387.89	-4.54%	₪995,329.91

External cash additions and artificial portfolio scaling are not investment profits. Conditional fund allocations change modeled coverage; their value is not added P&L. Lot/account representation cases leave equity unchanged. Bond shocks are mark sensitivities, not an assumed basis-point/duration calculation.

How formula quality was scored

The rubric was frozen before executing the sweeps. Half the grade comes from executed structural behavior; half is judgmental design quality. This identifies arithmetic and design weaknesses, not the formula most likely to predict future losses.

Behavioral comparisons	F0	F1	F2	F3	F4
Scale invariance	100/100	100/100	100/100	100/100	100/100
Same-security lot split	0/100	100/100	100/100	100/100	100/100
Cash-sale monotonicity	195/200	200/200	200/200	200/200	200/200
Borrowing purchase monotonicity	100/100	100/100	100/100	100/100	100/100
Equivalent account repartition	0/100	100/100	100/100	100/100	100/100

Each behavioral criterion earns 10 × its pass fraction. The cash criterion includes 100 equity-sale and 100 money-fund-redemption comparisons. Leverage purchases hold equity fixed and spend positive account cash before borrowing. Lot tests preserve each issuer, quantity and currency exposure; repartitions retain account currency and horizon and create no debt.

Quality component	F0	F1	F2	F3	F4
Interpretability /20	10	15	20	18	17
Evidence and coverage /20	9	13	14	14	14
Risk-channel breadth /10	6	7	9	9	9
Executed behavior /50	29.75	50.00	50.00	50.00	50.00
Total quality /100	54.75	85.00	93.00	91.00	90.00

What the static grade rewards

Interpretability allocates 5 points each to meaningful output units, explicit coefficients, attribution, and separating downside from suitability. Evidence allocates 5 each to a dated input snapshot, signed cash/FX handling, empirical calibration, and unsupported/missing inputs. Breadth allocates 2 each to equities, bonds, FX, funding, and issuer concentration. The full subcriterion allocations are retained in methodology_frozen.json.

Every model receives zero for empirical calibration. None has been fitted or tested on a chronological holdout of observed losses. All receive only partial funding credit: spending needs, settlement restrictions, verified margin requirements and pension liquidity are absent. Stale marks and missing fund allocations remain material regardless of the quality score.

Sensitivity and limits of the ranking

Reweighting behavioral versus static grades from 30/70 to 70/30 preserves the order F2, F3, F4, F1, F0. This checks that one category weight does not reverse the result. It does not prove robustness to different subjective subgrades or shock coefficients. F2’s two-point lead over F3 is a design preference, not statistical significance.

The 100 variants in each family are deliberately related. More variants improve coverage of implementation behavior, but do not supply 100 independent financial outcomes or confidence intervals. Prospective loss validation still requires dated, adjusted price/FX histories and a frozen scoring-to-outcome contract.

Financial interpretation and missing coverage

Modeled allocation	Value in ILS	Share of modeled equity
Individual equities	₪365,547.43	35.06%
Bonds	₪71,184.53	6.83%
ILS money-market fund	₪452,557.50	43.40%
Cash	₪153,428.33	14.71%
USD exposure proxy (overlaps above)	₪551,340.65	52.88%

The USD exposure estimate follows saved currency classifications and each stock’s quote-currency proxy. It is not verified revenue-currency look-through. A 20% USD decline alone reduces modeled equity by about 10.58% before any local price move. Fund liquidity, bank deposit risk and inflation risk remain separate from the market-shock score.

Account risk under each formula

Account	Current F0	F1	F2	F3	F4
ETrade ...1048	47	38	90	40	75
Schwab ...5122	42	41	100	45	83
Schwab ...2873	34	26	71	29	58
Psagot ...7641	25	3	2	1	2

The low market-risk estimate for the ILS money-fund account should not be read as zero risk or a personal allocation recommendation. Conversely, scores near 100 in USD-equity accounts reflect the chosen 50%-equity/20%-FX shock and score scaling, not a prediction that the account will fail.

100 conditional allocations for the unmodeled funds

Keeping the recorded ₪3,202,128.89 fund balance fixed, the review sweeps an assumed 0%-100% unhedged USD broad-equity share; the remainder is assumed ILS money market. Across 100 cases, whole-wealth F2 ranges from **14.17 to 91.11**, equivalent to **7.08%-45.56%** loss under the selected shocks.

This is a conditional illustration, not a bound on actual fund risk: the real allocation may lie outside those assumptions, include other currencies or instruments, or face restrictions. No whole-wealth baseline risk is assigned. Obtain dated fund holdings/exposure reports and refresh the balances before adopting one.

Bonds need instrument-specific follow-up. The stored name of 121899CH9 suggests a 2045 maturity, but issuer, duration, credit quality and clean/dirty price basis were not independently established. Sparse constant marks produce a zero range in the current formula. A generic 15% stress or assumed rate factor cannot certify its true downside. The 100 bond-mark sensitivities separately span 0.2%-20% declines.

Worked example, evidence, and reproduction

Joint stress arithmetic on the frozen investment snapshot

Selected stress component	Illustrative loss
USD individual stocks: (0.50 × 0.80) - 1	₪219,328.46
USD government bond: (0.88 × 0.80) - 1	₪11,201.48
USD unclassified bond: (0.85 × 0.80) - 1	₪10,669.34
USD cash: 20% FX loss	₪22,921.74
ILS money market:1% mark loss	₪4,525.57
Total	₪268,646.59

The stock loss is 60%, rather than 70%, because local-price and FX returns compound. Total loss / ₪1,042,717.80 =25.7641%; ×200 gives 51.5281. The 100 joint-sweep endpoint on page 5 is more severe (USD -25%) than this formula's reference FX shock (-20%), so its realized illustrative loss can exceed the reference catalogue result.

Primary methodology references

1. SEC Investor.gov: Beginners' Guide to Asset Allocation, Diversification, and Rebalancing
Allocation, diversification, goals and horizon as separate suitability considerations; no endorsement of app scores.

2. SEC Investor.gov: Money Market Funds
General comparative risk: liquid short-duration debt funds differ from equities and are not risk-free. US fund rules do not verify Israeli product terms.

3. SEC Investor Bulletin: Understanding Margin Accounts
Borrowing amplifies equity losses and can cause forced liquidation; not verified broker margin permissions.

4. BIS Basel Committee: Stress testing principles
Document objectives, assumptions, methodology and limitations. Banking principles used as methodological reference only.

5. SEC Investor.gov: Bonds - FAQs
Bond issuer/credit, interest-rate and liquidity risk; does not establish ratings, durations or clean/dirty prices of held bonds.

Retrieved 13/11/2026. These sources support risk mechanisms and documenting assumptions. They do not endorse the formulas, shock sizes, scores, fund classifications or broker permissions in this report. US money-market guidance is general context, not verification of Israeli fund 5139324. No web search snippet is used as a current valuation.

Reproduction and scope of verification

Run from the repository root:
./.venv/bin/python .runtime/risk-formula-review-2026-11-13/prepare.py
./.venv/bin/python .runtime/risk-formula-review-2026-11-13/run_review.py
./.venv/bin/python .runtime/risk-formula-review-2026-11-13/build_report.py

The evidence directory retains input.json, snapshot.json, current_model.js, candidate_formulas.py, protocol.json, methodology_frozen.json, value_audit.json, financial_examples.json, results.json and simulations.csv. Capture uses read-only database connections and rejects a moving source. Research scripts and output remain local; no formula was installed, no ledger corrected, no trade submitted, and no deployment performed.

Verification includes exact-model execution, source-to-ledger reconciliation, structural simulation invariants and 23 raw-source arithmetic checks by the dedicated financial analyst. That analyst authored the candidate methods; the raw-source review is a cross-check of the coordinator's reconciliation and simulations, not a separate independent audit of the analyst's models. No historical forecast validation, broker-statement reconciliation, or live-browser acceptance test is claimed.